BHAVIK ACHOTANI

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EDUCATION

Thadomal Shahani Engineering College (University of Mumbai) **Mumbai, India**
Bachelor of Engineering (B.E.) in Information Technology | **CGPA: 8.08 / 10.0** | Academic Standing: 100% Course Clearance (Zero Backlogs) **Expected May 2027**
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, Database Management Systems (DBMS), Object-Oriented Programming, Operating Systems.

RESEARCH PUBLICATIONS & PREPRINTS

- **Achotani, B.** (2026). "NeuroSeg: A Bi-Hemispheric Spectral-Attention U-Net Workspace with Signed Distance Boundary Supervision, Bayesian Uncertainty Calibration, and Cross-Modal Neuro-Hematology Fusion." *Zenodo Research Archive*. DOI: 10.5281/zenodo.24046115 [IEEE Two-Column Format].
- **Achotani, B.** (2026). "DhanSathi: Offline-First Multilingual Agricultural Intelligence and Deterministic RAG Architecture on Low-Resource Mobile Edges." *SSRN Electronic Journal*. Available at SSRN: 7670799.
- **Achotani, B.** (2026). "Quantx: High-Concurrency Transactional State-Locking and Vectorized Backtesting in Low-Latency Financial Simulators." *SSRN Electronic Journal*. Available at SSRN: 7670718.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, Kotlin, Solidity, C, SQL
AI, ML & Deep Learning: PyTorch, Attention U-Net, Bayesian ML, Scikit-learn, XGBoost, Pandas, NumPy, Groq LLaMA 3.3, RAG, Circom ZK-SNARKs
Web, Mobile & Backend: React.js, Node.js, Express.js, FastAPI, Android (Jetpack Compose), Streamlit, Plotly, REST APIs, WebSockets
Systems, Databases & Tools: Linux/Bash, Git, Docker, MongoDB, MySQL, Redis, Firebase, Room DB, Hardhat, Ethers.js

WORK EXPERIENCE

Context AI — ***Software Engineering Intern*** **Nov 2025 – Jan 2026**
- Engineered and maintained full-stack web application modules using React.js and Node.js, reducing front-end page load latency by 28% across client workflows.
- Designed RESTful APIs and optimized MongoDB database schemas with query indexing and validation, improving backend response consistency under concurrent user load.
- Collaborated with the core engineering team to debug production bottlenecks, implement security best practices, and streamline API payload serialization.

Indian Institute of Computing and Technology (IICT) — ***AI & Machine Learning Intern*** **Dec 2026 – Jan 2027 (Upcoming)**
- Selected for the competitive Winter AI & Machine Learning Internship Program (commencing Dec 10, 2026) focusing on deep learning model training, evaluation pipelines, and applied AI deployment.

PROJECTS

NeuroSeg — Multimodal Brain MRI PACS & Liquid-Biopsy AI | GitHub | Zenodo | Demo *PyTorch, FastAPI, React 19, TypeScript*
- Architected a 4-tier clinical neuro-oncology PACS uniting a 4-stage Spectral-Transformer Attention U-Net (2D FFT high-pass gates + contralateral bi-hemispheric cross-attention) with a 6-layer MRI viewer and live scan dropzone.
- Formulated a GPU morphological Signed Distance Map (SDM) loss and Bayesian Monte Carlo dropout engine (T=8 passes), boosting Dice Similarity from 83.10% to 91.84%, cutting HD95 error by 60.4% (3.42mm), and achieving 0.0192 ECE.
- Trained a Cross-Modal Gated Fusion Network on 12,000 multimodal patient records to fuse 3D MRI volumetry with a 12-biomarker blood & Serum GFAP panel (94.33% 4-class differential accuracy, 0.964 ROC-AUC); published IEEE paper on Zenodo.

DhanSathi — AI Smart Farming & Financial Companion | GitHub | SSRN | Demo *Kotlin, FastAPI, Groq LLaMA 3.3, Firebase*
- Built a full-stack multilingual Android app (English, Hindi, Marathi) delivering AI crop advisory, government scheme matching, and financial tracking for Indian farmers.
- Developed a FastAPI backend integrating Groq's LLaMA 3.3 70B model with <850ms inference latency for yield prediction, cost/profit estimation, and advisory across 10 major Maharashtra crops.
- Implemented a bilingual AI chatbot and voice-based transaction parser using Room DB for offline-first storage and Firebase for cloud synchronization; authored technical preprint on SSRN.

Quantx — Stock Market & AI Backtesting Simulator | GitHub | SSRN | Demo *React.js, Node.js, Python, Redis, WebSockets*
- Architected a high-concurrency trading simulation engine for Indian equities and F&O data, featuring a state-locked transactional wallet managing ₹1L–₹50L virtual capital with zero ledger drift.
- Engineered a vectorized AI strategy backtesting module in Python and Pandas to process 5+ years of historical OHLCV time-series datasets and compute Sharpe ratio and maximum drawdown.
- Constructed low-latency WebSocket market streaming with Redis pub/sub caching, cutting order-book latency to <45ms while modeling exchange slippage and STT tax calculations; authored preprint on SSRN.

Telecom Customer Churn Prediction & Analytics Dashboard | GitHub | Demo *Python, XGBoost, Scikit-learn, Streamlit, Plotly*
- Trained and optimized an XGBoost classification model on 7,000+ telecom customer records (contract, billing, tenure, usage metrics), achieving 84.2% ROC-AUC and 82% F1-score in flagging high-risk churn.
- Designed an interactive 4-tab Streamlit analytics dashboard with Plotly visualizations; deployed across a local network with IP-based multi-device access for cross-platform model evaluation.
- Integrated SHAP (SHapley Additive exPlanations) tree interpretability into the inference pipeline, attributing feature-level risk contributions (month-to-month contracts +32.4%) to guide retention strategies.

InsureChain — Decentralized ZK Insurance Platform | GitHub | Demo *Solidity, Circom ZK-SNARKs, Hardhat, Node.js, MongoDB*
- Engineered a decentralized insurance protocol with Circom ZK-SNARKs (<1.5s proof generation), enabling policyholders to cryptographically prove coverage eligibility without exposing sensitive data on-chain.
- Authored Solidity smart contracts on EVM with automated Groth16 on-chain verifier integration, optimizing BN254 pairing check gas consumption down to 234,180 gas per underwriting audit.
- Built a reactive front-end dashboard with client-side SnarkJS witness generation and MetaMask integration, ensuring medical health parameters remain strictly confidential in local memory.

CERTIFICATIONS

IBM — Generative AI Foundational Models for NLP & Language Understanding — September 2026
IBM — Deep Learning and Reinforcement Learning — September 2026
IBM — Introduction to Artificial Intelligence — August 2026